

Ex.no:3	DATA CLEANING

Aim:**Algorithm:**

Check Missing Values:

1. Start.
2. Import the pandas library.
3. Load the Life Expectancy dataset into a DataFrame (df).
4. Display dataset information using df.info().
5. Check missing values using df.isnull().
6. Count missing values using df.isnull().sum().
7. Check missing values using df.isna().
8. Check non-missing values using df.notnull().
9. Display the results.

Drop Missing Values

1. Start.
2. Import the pandas library.
3. Load the Life Expectancy dataset into a DataFrame (df).
4. Drop missing rows using dropna().
5. Drop rows (axis=0) or columns (axis=1).
6. Use how='any' to remove rows/columns containing any missing value.
7. Use how='all' to remove rows/columns only if all values are missing.
8. Use thresh=N to keep rows/columns with at least N non-missing values.
9. Use subset=['column_name'] to drop rows based on specific columns.
10. Use inplace=True to modify the original DataFrame.
11. Display the cleaned DataFrame.

Program:

```
import pandas as pd
import os
import kagglehub

# Load the Life Expectancy dataset

path = "/content"

print("Dataset path:", path)
```

```
print("\nFiles in dataset:")
print(os.listdir(path))

# Find CSV file automatically

csv_files = [f for f in os.listdir(path) if f.lower().endswith(".csv")]

if not csv_files:
    raise FileNotFoundError("No CSV file found in the dataset path.")

csv_file = csv_files[0]

print("\nCSV file:", csv_file)

# Load dataset

df = pd.read_csv(
    os.path.join(path, csv_file),
    encoding="latin1",
    low_memory=False
)

# Display selected Life Expectancy dataset columns

data = {

    "Country": df["Country"].head(5).tolist(),

    "Year": df["Year"].head(5).tolist(),

    "Status": df["Status"].head(5).tolist(),

    "Life_Expectancy": df["Life expectancy"].head(5).tolist(),

    "Adult_Mortality": df["Adult Mortality"].head(5).tolist(),

    "GDP": df["GDP"].head(5).tolist(),

    "Schooling": df["Schooling"].head(5).tolist()

}

life_expectancy_df = pd.DataFrame(data)

print("Original DataFrame:")

print(life_expectancy_df)

print("\n1. First 5 rows (head):")

print(life_expectancy_df.head())

print("\n2. Last 2 rows (tail):")

print(life_expectancy_df.tail(2))
```

```
print("\n3. Shape (rows, columns):")

print(life_expectancy_df.shape)

print("\n4. Column Names:")

print(life_expectancy_df.columns)

print("\n5. DataFrame Information:")

life_expectancy_df.info()

print("\n6. Statistical Summary:")

print(life_expectancy_df.describe())

print("\n7. High life expectancy records:")

print(
    df[df["Life expectancy"] > 75][
        [
            "Country",
            "Year",
            "Status",
            "Life expectancy",
            "GDP",
            "Schooling"
        ]
    ].head()
)

print("\n8. Records with life expectancy less than 60:")

print(
    df[df["Life expectancy"] < 60][
        [
            "Country",
            "Year",
            "Status",
            "Life expectancy",
```

```

        "GDP",
        "Schooling"
    ]

    ].head()

)

```

Output:

```

Path      Country Year   Status Life expectancy Adult Mortality \
0  Afghanistan  2015   Developing          65.0          263.0
1  Afghanistan  2014   Developing          59.9          271.0
2  Afghanistan  2013   Developing          59.9          268.0
3  Afghanistan  2012   Developing          59.5          272.0
4  Afghanistan  2011   Developing          59.2          275.0

    infant deaths  Alcohol  percentage expenditure  Hepatitis B  Measles
... \
0              62      0.01          71.279624          65.0      1154
...
1              64      0.01          73.523582          62.0       492
...
2              66      0.01          73.219243          64.0       430
...
3              69      0.01          78.184215          67.0      2787
...
4              71      0.01          7.097109          68.0      3013
...

    Polio  Total expenditure  Diphtheria  HIV/AIDS          GDP
Population \
0      6.0              8.16          65.0          0.1  584.259210
33736494.0
1      58.0              8.18          62.0          0.1  612.696514
327582.0
2      62.0              8.13          64.0          0.1  631.744976
31731688.0
3      67.0              8.52          67.0          0.1  669.959000
3696958.0
4      68.0              7.87          68.0          0.1   63.537231
2978599.0

    thinness 1-19 years  thinness 5-9 years  Income composition of
resources \
0              17.2              17.3
0.479
1              17.5              17.5
0.476
2              17.7              17.7
0.470
3              17.9              18.0
0.463

```

4 18.2 18.2
0.454

Schooling
0 10.1
1 10.0
2 9.9
3 9.8
4 9.5

[5 rows x 22 columns]
(2938, 22)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2938 entries, 0 to 2937
Data columns (total 22 columns):

#	Column	Non-Null Count	Dtype
0	Country	2938 non-null	object
1	Year	2938 non-null	int64
2	Status	2938 non-null	object
3	Life expectancy	2928 non-null	float64
4	Adult Mortality	2928 non-null	float64
5	infant deaths	2938 non-null	int64
6	Alcohol	2744 non-null	float64
7	percentage expenditure	2938 non-null	float64
8	Hepatitis B	2385 non-null	float64
9	Measles	2938 non-null	int64
10	BMI	2904 non-null	float64
11	under-five deaths	2938 non-null	int64
12	Polio	2919 non-null	float64
13	Total expenditure	2712 non-null	float64
14	Diphtheria	2919 non-null	float64
15	HIV/AIDS	2938 non-null	float64
16	GDP	2490 non-null	float64
17	Population	2286 non-null	float64
18	thinness 1-19 years	2904 non-null	float64
19	thinness 5-9 years	2904 non-null	float64
20	Income composition of resources	2771 non-null	float64
21	Schooling	2775 non-null	float64

dtypes: float64(16), int64(4), object(2)

memory usage: 505.1+ KB

Missing Values

Country	0
Year	0
Status	0
Life expectancy	10
Adult Mortality	10
infant deaths	0
Alcohol	194
percentage expenditure	0
Hepatitis B	553
Measles	0
BMI	34
under-five deaths	0
Polio	19
Total expenditure	226
Diphtheria	19

HIV/AIDS	0
GDP	448
Population	652
thinness 1-19 years	34
thinness 5-9 years	34
Income composition of resources	167
Schooling	163
dtype: int64	

Missing Values After Handling

Country	0
Year	0
Status	0
Life expectancy	0
Adult Mortality	0
infant deaths	0
Alcohol	0
percentage expenditure	0
Hepatitis B	0
Measles	0
BMI	0
under-five deaths	0
Polio	0
Total expenditure	0
Diphtheria	0
HIV/AIDS	0
GDP	0
Population	0
thinness 1-19 years	0
thinness 5-9 years	0
Income composition of resources	0
Schooling	0
dtype: int64	
Duplicate Rows : 0	
After Removal : (2938, 22)	

Life Expectancy Outliers

	Country	Year	Status	Life expectancy	Adult Mortality \
1127	Haiti	2010	Developing	36.3	682.0
1484	Lesotho	2005	Developing	44.5	675.0
1582	Malawi	2003	Developing	44.6	613.0
1583	Malawi	2002	Developing	44.0	67.0
1584	Malawi	2001	Developing	43.5	599.0

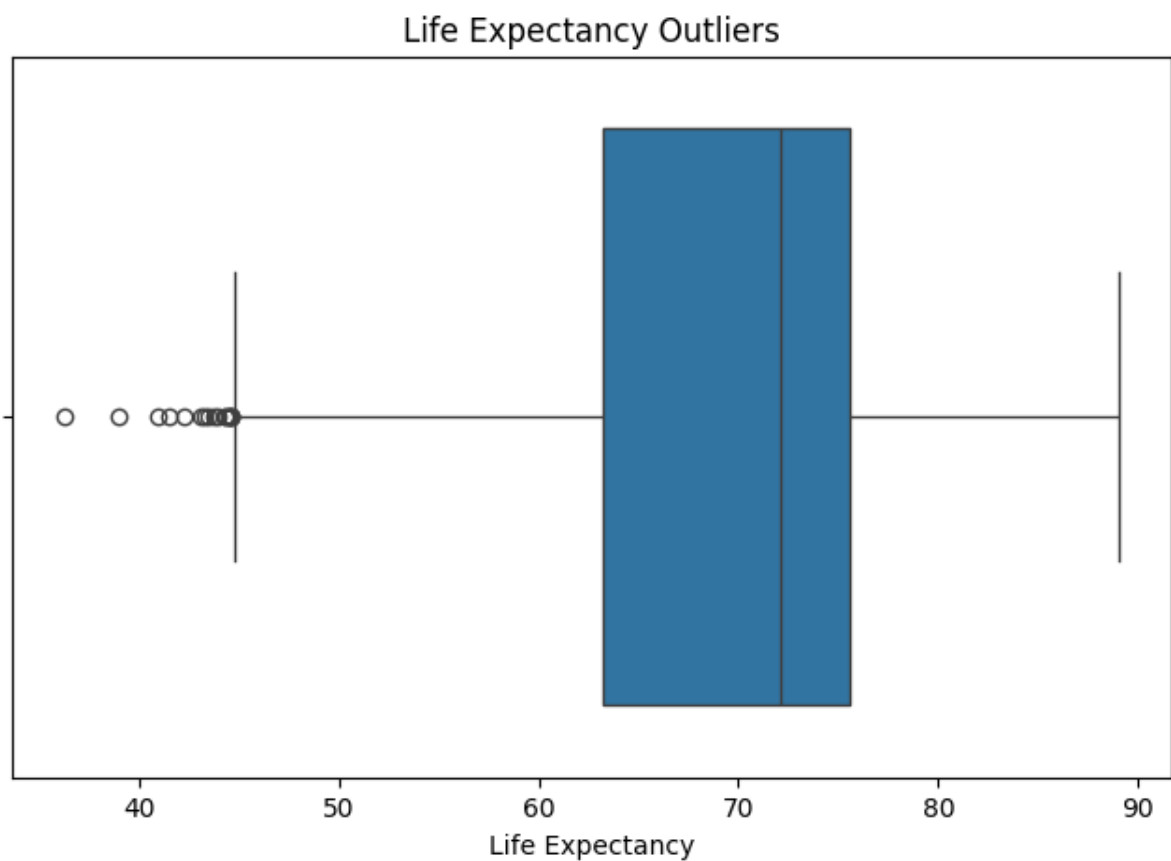
	infant deaths	Alcohol	percentage expenditure	Hepatitis B
Measles \				
1127	23	5.76	36.292918	92.0
0				
1484	5	2.67	57.903698	87.0
0				
1582	43	1.08	4.375316	84.0
167				
1583	46	1.10	3.885395	64.0
92				
1584	48	1.15	12.797606	92.0
150				

	...	Polio	Total expenditure	Diphtheria	HIV/AIDS	GDP
\						
1127	...	66.0	8.90	66.0	1.9	662.279518
1484	...	88.0	6.30	89.0	34.8	862.946312
1582	...	85.0	6.35	84.0	24.2	26.152517
1583	...	79.0	4.82	64.0	24.7	29.979898
1584	...	86.0	5.70	9.0	25.1	146.761540

	Population	thinness 1-19 years	thinness 5-9 years	\
1127	9999617.0	4.0	4.0	
1484	1949543.0	9.3	9.2	
1582	12336687.0	7.6	7.5	
1583	1213711.0	7.7	7.6	
1584	11695863.0	7.9	7.7	

	Income composition of resources	Schooling
1127	0.470	8.6
1484	0.437	10.7
1582	0.362	10.3
1583	0.388	10.4
1584	0.387	10.1

[5 rows x 22 columns]



Normalized Life Expectancy

	Life expectancy
0	0.544592
1	0.447818
2	0.447818
3	0.440228
4	0.434535

```
Encoded Status
Status
```

```
0      1
1      1
2      1
3      1
4      1
```

```
Encoded Country
Country
```

```
0      0
1      0
2      0
3      0
4      0
```

Cleaned dataset saved successfully.

Final Dataset Shape : (2938, 22)

Final Missing Values:

```
Country      0
Year         0
Status       0
Life expectancy  0
Adult Mortality  0
infant deaths  0
Alcohol       0
percentage expenditure  0
Hepatitis B   0
Measles       0
BMI           0
under-five deaths  0
Polio         0
Total expenditure  0
Diphtheria    0
HIV/AIDS      0
GDP           0
Population    0
thinness 1-19 years  0
thinness 5-9 years  0
Income composition of resources  0
Schooling     0
dtype: int64
```

Result: